

C:\Users\srika\PycharmProjects\PythonProject\.venv\Scripts\python.exe "C:\Users\srika\PycharmProjects\PythonProject\main.py"

Word	Root	Affix(es)	Type	Normalized
analyzing	analyze	-ing	Inflectional	analyze
analysis	analyze	-sis (noun-forming)	Derivational	analyze
analytical	analyze	-tical (adj-forming)	Derivational	analyze

Detailed Structured Report:

Original Word : analyzing
Root : analyze
Affix(es) : -ing
Transformation Type : Inflectional
Semantic Note : Present participle / progressive form of the verb 'analyze'.
Normalized (Index) Form : analyze

Original Word : analysis
Root : analyze
Affix(es) : -sis (noun-forming)
Transformation Type : Derivational
Semantic Note : Noun form derived from the verb 'analyze' (verb -> noun).
Normalized (Index) Form : analyze

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C:\Users\srika\PycharmProjects\PythonProject\.venv\Scripts\python.exe "C:\Users\srika\PycharmProjects\PythonProject\NLP\C02_AT2\NLP 2.py"
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Word	Prefix	Root	Suffix	Category	Polarity

disagree	dis-	agree	None	Derivational (prefixation)	Negative
agreement	None	agree	-ment	Derivational (verb -> noun)	Neutral/Positive
agreeable	None	agree	-able	Derivational (verb -> adjective)	Positive

Comprehensive Output:

Original Word	: disagree
Prefix	: dis-
Root Word	: agree
Suffix	: None
Transformation Category	: Derivational (prefixation)
Semantic Interpretation	: Negation of the root meaning: 'agree' -> 'not agree' (opposite polarity, sentiment-reversing).
Normalized Base Form	: agree
Sentiment Polarity	: Negative

Original Word	: agreement
Prefix	: None
Root Word	: agree
Suffix	: -ment
Transformation Category	: Derivational (verb -> noun)
Semantic Interpretation	: Converts the verb 'agree' into a noun denoting the state/result of agreeing (neutral polarity carried from root).
Normalized Base Form	: agree
Sentiment Polarity	: Neutral/Positive

Original Word	: agreeable
Prefix	: None
Root Word	: agree
Suffix	: -able

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C:\Users\srika\PycharmProjects\PythonProject\.venv\Scripts\python.exe "C:\Users\srika\PycharmProjects\PythonProject\NLP\C02_AT2\NLP_3.py"
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Word	Root	Affix(es)	Deriv.Lvl	POS
govern	govern	None (base form)	0	Verb
government	govern	-ment	1	Noun (institution / act of governing)
governance	govern	-ance	1	Noun (system / process of governing)

Structured Output for Topic Modeling / Clustering:

Original Word	: govern
Root Form	: govern
Detected Affix(es)	: None (base form)
Derivational Level	: 0
Part of Speech	: Verb
Normalized Representation	: govern
Original Word	: government
Root Form	: govern
Detected Affix(es)	: -ment
Derivational Level	: 1
Part of Speech	: Noun (institution / act of governing)
Normalized Representation	: govern
Original Word	: governance
Root Form	: govern
Detected Affix(es)	: -ance
Derivational Level	: 1
Part of Speech	: Noun (system / process of governing)
Normalized Representation	: govern

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Root Word      : active
Suffix         : -ate
Derivational Sequence : active -> activate (-ate: adjective -> verb)
Word-Class Change  : Adjective -> Verb
Normalized Base Form : active
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Original Word   : activation
Prefix         : None
Root Word      : active
Suffix         : -ate + -ion
Derivational Sequence : active -> activate -> activation (-ate: adjective -> verb, -ion: verb -> noun)
Word-Class Change  : Adjective -> Verb -> Noun
Normalized Base Form : active
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Original Word   : reactivation
Prefix         : re-
Root Word      : active
Suffix         : -ate + -ion
Derivational Sequence : active -> activate -> activation -> reactivation (re-: repetition prefix added to noun)
Word-Class Change  : Adjective -> Verb -> Noun (+ 're-' repetition meaning)
Normalized Base Form : active
```

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Final Parsed Representation:
activate      -> [active]+-ate (normalized: active)
activation    -> [active]+-ate + -ion (normalized: active)
reactivation  -> re-[active]+-ate + -ion (normalized: active)
```

Structured Output for Search Optimization / Information Retrieval:

Original Word : create
Identified Suffix : None (base form)
Grammatical Category : Base form (infinitive / present tense, all persons except 3rd singular)
Extracted Root : create
Normalized Base Form : create
Final Normalized Representation : create

Original Word : creates
Identified Suffix : -s
Grammatical Category : Third-person singular present tense
Extracted Root : create
Normalized Base Form : create
Final Normalized Representation : create

Original Word : creating
Identified Suffix : -ing
Grammatical Category : Present participle / gerund (continuous aspect)
Extracted Root : create
Normalized Base Form : create
Final Normalized Representation : create

Normalization Group (Base Form -> Variants Mapped):
"create" <= ['create', 'creates', 'creating']

Process finished with exit code 0